

An Implementation of a Natural Language Processing Model with a Custom-Built Chemical Patent Database for OLEDs

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In recent studies, machine learning has been implemented to assist the discovery of materials for more efficient and durable OLEDs. However, a molecular design process requires researchers to choose molecular groups to investigate, which limits this process to the researchers' knowledge of different molecular groups. In order to help researchers expand their molecular design space for OLEDs, a patent database can be used to provide additional learnings. A patent is often the first publication medium where researchers, especially in industry, publish new discoveries. Thus, many studies have used patent databases to extract chemical knowledge. However, to the best of our knowledge, such studies have been limited to patents that are for biotech/pharmaceutical applications, commercially available at cost, or not maintained currently.^[1] Therefore, in this work, we built a chemical patent database for OLEDs that provides molecular structure information from the patents, and implemented a pre-trained natural language processing (NLP) model to generate a list of chemical materials patents for each layer of an OLED.

Fig. 1(a) shows a workflow to build a chemical patent database for OLEDs. We used the United States Patent and Trademark Office (USPTO) patent bulk data, which includes XML files containing text content of patents, and other files containing non-text content such as TIF image files, MDL molfiles and ChemDraw CDX binary files, if applicable. For the non-text content, i.e., molecular structure information in the patents, we utilized a ChemAxon software and an in-house code to convert the files to SMILES (simplified molecular-input line-entry system) strings and post-process resolving issues that frequently occur when using general tools.

Then, for the textual content, we implemented a pre-trained natural language processing (NLP) model, PatentSBERTa, which is based on S-BERT (sentence bidirectional encoder representations from transformers) and RoBERTa (robustly optimized BERT pretraining approach) and is fine-tuned for patent claims.^[2] Fig. 1(b) shows an exemplary t-SNE (t-distributed stochastic neighbor embedding) map of PatentSBERTa-based embedding vectors of text content in a patent. Among numerous patents under the same IPC/CPC (international patent classification/cooperative patent classification), patents with similar content can be found based on the cosine similarity of the embedding vectors. With this approach, we were able to generate lists of chemical materials patents for each layer of an OLED, depending on their similarity with patents of interest.

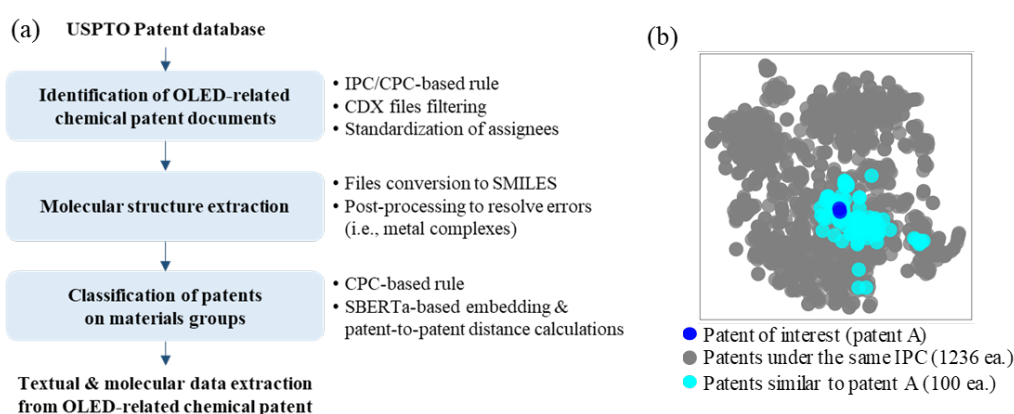


Fig. 1. (a) A workflow to build a chemical patent database for OLEDs; (b) An exemplary t-SNE map of embedding vectors of patent claims

References

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